An Integrated Explainable AI Platform for Academic Guidance, Careers, and Wellbeing

R26-IT-073

Project Proposal Report

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Match 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Many students struggle to make informed academic and career decisions due to the lack of direct access to experienced professionals and verified mentors. Although several online platforms provide career advice and networking opportunities, many of them lack proper mentor verification, structured interaction, and a secure environment specifically designed for students. This creates a gap where students may receive unreliable guidance or have difficulty finding trustworthy mentors who can provide meaningful career insights.

The identified research gap lies in the absence of a **secure, verified, and structured mentorship system** that connects students with qualified mentors while ensuring authenticity and quality of interactions. Existing systems rarely combine mentor verification, real-time communication, and achievement recognition within a single platform.

To address this issue, this project proposes a **Professional Mentorship and Alumni Network module** that enables students to connect with verified mentors such as alumni, university seniors, and industry professionals. The system will include a **mentor verification mechanism using OCR-based document validation and interview assessment**, ensuring that only credible mentors are approved. Additionally, a **badge assignment system (Bronze, Silver, Gold)** will recognize mentor contributions and encourage active participation. The platform will also support **real-time communication between mentors and students**, enabling guidance, knowledge sharing, and career discussions.

The system will be developed using modern technologies including **React Native for mobile application development, Firebase for real-time chat integration, Node.js for backend services, and OCR-based verification mechanisms** for validating mentor credentials.

The expected outcomes include improved access to reliable mentorship, better student decision-making regarding academic and career paths, and stronger connections between universities and alumni networks. Socially, the system promotes knowledge sharing and professional guidance among students. Commercially, it can be expanded into a scalable mentorship platform for educational institutions, training organizations, and professional communities.

Keywords: Mentorship Platform, Alumni Network, Mentor Verification, Real-Time Communication, Student Career Guidance.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my research supervisor, **Ms. Thilini Jayalath, Senior Lecturer**, for her valuable guidance, constructive feedback, and continuous support throughout the preparation of this research proposal. Her expertise, encouragement, and insightful advice greatly contributed to improving the quality and direction of this work.

I would also like to extend my appreciation to my co-supervisor, **Ms. Jenny Kishara, Senior Lecturer**, for her helpful suggestions and guidance during the research planning process. Her support and recommendations were highly valuable in shaping and refining this study.

I am also grateful to the **Faculty of Computing at the Sri Lanka Institute of Information Technology (SLIIT)** for providing the academic resources, facilities, and a supportive learning environment necessary for conducting this research.

Finally, I would like to express my heartfelt thanks to my friends and family for their continuous encouragement, patience, and moral support throughout the preparation of this proposal. Their motivation played an important role in helping me successfully complete this work.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form / Meaning
AI	Artificial Intelligence
API	Application Programming Interface
OCR	Optical Character Recognition
UI	User Interface
UX	User Experience
SUS	System Usability Scale
JWT	JSON Web Token (used for authentication)
Firebase	Google's real-time backend service (used for chat and temporary storage)
Supabase	Open-source PostgreSQL-based backend-as-a-service (database & authentication)
Expo	Open-source platform for React Native development and deployment
CV	Curriculum Vitae
SDG	Sustainable Development Goal (SDG 4 & 8 referenced in background)
PDPA	Personal Data Protection Act (Sri Lanka)
IRB	Institutional Review Board (or Ethics Committee)
NLP	Natural Language Processing (used in basic interview scoring)
TF-IDF	Term Frequency-Inverse Document Frequency (used in matching engine)
APK	Android Package (for app distribution during testing)

1. INTRODUCTION

1.1 Background and Context

In Sri Lanka, thousands of undergraduates and post-A/L students face significant challenges in accessing reliable career and academic guidance due to the absence of a trusted, easily accessible mentorship platform. Most students depend on informal networks, limited university career centres, or generic online tools, which often lead to mismatched career choices, prolonged underemployment, and delayed professional growth. This lack of structured mentorship creates a critical gap between academic qualifications and industry expectations, leaving many graduates unprepared for the job market.

This problem directly affects over 80,000 university students and tens of thousands of post-A/L students annually, including those at institutions such as SLIIT, University of Colombo, University of Moratuwa, and University of Peradeniya. Without access to verified mentors, students make uninformed decisions that contribute to higher dropout rates, extended periods of unemployment, and wasted academic potential. Reliable mentorship is essential because it bridges the gap between theoretical knowledge and practical industry needs, enabling students to build confidence, develop relevant skills, and establish professional networks early in their careers.

According to the Department of Census and Statistics Labour Force Survey (Third Quarter 2025), the national unemployment rate stands at 4.3%, while youth unemployment (ages 15–24) remains persistently high at 19.2%. Unemployment among individuals with GCE A/L qualifications and above reached 7.0% in the same period. Globally, research from the National Bureau of Economic Research indicates that students with access to mentors are 14% more likely to remain in university and 13% more likely to graduate, highlighting how weak mentorship in Sri Lanka exacerbates the education–job market mismatch.

Existing solutions, such as university career guidance units, LinkedIn, and international platforms like MentorCruise or local social media groups, suffer from major limitations. Most rely on manual or absent verification processes, lack privacy safeguards, and offer no structured interview or document validation mechanisms. These systems also fail to provide real-time secure communication or trust indicators such as verified badges, resulting in low student confidence and limited adoption, particularly in the Sri Lankan context.

To address these challenges, the proposed **Professional Mentorship and Alumni Network** component introduces a secure, automated platform featuring mentor–student matching based on interests, degrees, and career goals; multi-layered verification through CV and certificate uploads with OCR-based extraction combined with structured interview scoring; secure real-time chat functionality; and mentor profiles with Bronze, Silver, and Gold verified badges. The system adopts a privacy-preserving approach with temporary encrypted storage and automatic deletion of raw documents, ensuring seamless integration with the broader AI platform.

This component aligns with **SDG 4 (Quality Education)** by promoting lifelong learning opportunities and **SDG 8 (Decent Work and Economic Growth)** by enhancing career readiness and productive employment. It also fits within SLIIT's **Intelligent Systems & Software Engineering** research cluster, incorporating AI-driven matching, OCR/NLP

techniques, real-time systems, and privacy engineering — areas actively pursued within the Department of Information Technology.

1.2 Literature Review

Recent scholarly research published between 2023 and 2026 on digital mentorship and alumni networks has increasingly focused on the application of artificial intelligence to improve mentor–mentee matching, verification processes, privacy protection, and real-time interaction. Scholars have shifted from traditional manual university-based mentoring systems toward scalable, AI-supported platforms capable of serving large student populations. This literature review synthesises ten key sources from 2024 to 2026, examining current research approaches, existing commercial products and services, and their critical limitations.

Several studies have proposed AI-powered algorithms for mentor–student matching. Wahrini et al. (2026) developed an adaptive mentoring module using machine learning techniques to deliver personalised career guidance, demonstrating measurable improvements in student readiness through skills-based pairing. Similarly, the 2025 IJIRT paper titled "Career Mentorship Platform Using AI-Powered Recommendations" employed Recurrent Neural Networks and cosine similarity to match mentors with students based on academic profiles and career goals. Another significant contribution from IJARSCT (2025) introduced an intelligent tutoring and mentorship platform that dynamically integrates mentor availability, expertise, and learner needs for more effective pairing.

For mentor verification, emerging research has incorporated Optical Character Recognition (OCR) technologies. The "Margdarshak: AI-powered virtual career mentor" system (2025) utilises OCR combined with natural language processing to analyse CVs and certificates for credibility assessment. Likewise, a 2025 IRJMETS study on "Smart AI Career Advisor with Skill-Based Mentorship Matching" integrated automated document analysis to validate mentor qualifications. Privacy-preserving mechanisms have also been addressed in recent works; an IEEE paper (2024) on "Enhancing Alumni Engagement Through Digital Platform" demonstrated the implementation of encrypted temporary storage and compliance with data privacy regulations, while the 2025 Demand Mentorship Platform research emphasised on-demand booking with minimal data retention. Real-time communication has been explored in "Steve: LLM Powered ChatBot for Career Progression" (arXiv, 2025), which showcased conversational AI for continuous mentorship support, and in an IJARCCE study (2025) on "Alumni Association Platform Using Machine Learning," which integrated chatbots for instant student query handling.

On the commercial front, several established platforms currently dominate the market. Qooper (2025–2026) offers AI-assisted matching, progress tracking, and guided mentoring sessions and is widely adopted in enterprise environments. MentorcliQ and Chronus provide administrator-led matching and detailed analytics, while Together focuses on

enterprise scalability through integration with tools such as Slack and Microsoft Teams. Student-oriented solutions include UStrive, which provides free alumni mentoring for undergraduates, along with general-purpose tools such as LinkedIn and university career portals. In the Sri Lankan context, most higher education institutions still rely on manual alumni networks or basic social media groups with very limited automation.

Despite these advancements, significant limitations persist across both research and commercial systems. Most platforms continue to depend on manual or email-only verification methods and lack automated OCR for CVs and certificates as well as structured interview scoring. None implement a complete multi-layered trust-chain that combines document OCR with interview evaluation, nor do they incorporate temporary encrypted storage with automatic deletion of raw files, resulting in serious privacy risks and permanent data retention. Badge systems for trust indication are either absent or overly simplistic, reducing student confidence. Furthermore, real-time chat features are rarely integrated with broader AI career guidance ecosystems, and Sri Lankan-specific customisation — such as support for local universities and A/L students — remains missing. Matching accuracy often suffers without hybrid human–AI verification, and scalability is constrained by reliance on manual administrative oversight. These gaps create low trust, slow adoption, and poor integration with academic and career AI tools — precisely the deficiencies that the proposed Professional Mentorship and Alumni Network component aims to overcome through OCR-based document verification, structured interview scoring, privacy-first design, Bronze/Silver/Gold badges, and seamless integration with the larger AI platform.

1.3 Research Gap

Despite the rapid development of digital mentorship and alumni platforms in recent years, a critical research gap persists in the creation of a fully automated, privacy-preserving, and trustworthy mentorship system specifically designed for Sri Lankan undergraduates and post-A/L students. What is clearly missing is a complete **multi-layered verification mechanism** that automatically validates mentors through document upload (CV and certificates) using Optical Character Recognition (OCR) combined with structured interview scoring, while maintaining strict privacy controls.

Current approaches are insufficient because they continue to rely heavily on manual verification, simple email checks, or no verification at all. Most existing systems fail to implement temporary encrypted storage with automatic deletion of raw documents, leading to serious privacy risks and permanent data retention. They also lack a tiered badge system (Bronze, Silver, Gold) that clearly communicates the level of mentor credibility to students. Furthermore, real-time communication and matching features are rarely customised for the Sri Lankan higher education context or seamlessly integrated with broader AI career guidance platforms. These shortcomings have been repeatedly highlighted in recent studies (Wahrini et al., 2026; Margdarshak system, 2025; IEEE Alumni Platform, 2024), resulting in low student

trust, slow adoption rates, high manual administrative workload, and persistent education–job market mismatches.

The proposed **Professional Mentorship and Alumni Network** component directly fills this gap with a novel and creative solution. It introduces the first Sri Lankan-focused mentorship platform that combines OCR-based document verification and structured interview scoring into a seamless multi-layered trust-chain, enforces strong privacy through temporary storage and auto-deletion of files, and includes a three-level verified badge system. By integrating human mentorship as a formal decision-support layer within the larger AI platform, this component provides a practical, scalable, and highly trustworthy ecosystem that no existing research or commercial system currently offers.

2. OBJECTIVE

2.1 Main Objective

The main objective of this component is to **design, develop, and implement a secure, automated Professional Mentorship and Alumni Network platform** that connects Sri Lankan undergraduates and post-A/L students with reliably verified mentors from universities and industry professionals.

The platform will achieve this through multi-layered automated verification (document upload with OCR-based CV and certificate validation combined with structured interview scoring), a tiered verified badge system (Bronze, Silver, Gold), secure real-time chat functionality, and seamless integration with the larger AI platform, while ensuring full privacy compliance through temporary encrypted storage and automatic deletion of sensitive documents. The measurable target is to attain at least **85% verification accuracy**, **90% student satisfaction** in usability testing, and complete privacy compliance.

2.2 Specific Objectives

Here are 5 precise objectives for your **Professional Mentorship and Alumni Network** component:

1. **To implement a privacy-preserving mentor authentication system** utilizing Zero-Knowledge Proof principles to successfully verify 100% of registered alumni and university seniors without exposing their sensitive personal data.
2. **To design a Context-Aware Trust Filter** that algorithmically matches school students with verified academic mentors, targeting a minimum user match-satisfaction and relevance score of 80% during beta testing.
3. **To develop an Institutional Identity Bridge** that integrates and cross-validates user credentials against at least two distinct educational databases to ensure the absolute legitimacy of the alumni network.
4. **To establish a secure communication module** capable of facilitating and safely logging a minimum of 500 privacy-compliant mentorship interactions (such as messages or session bookings) within the first three months of platform deployment.
5. **To quantify the reduction of information asymmetry** by conducting pre- and post-intervention assessments, aiming to measure a 40% statistically significant increase in students' confidence regarding their academic and career pathway decisions.

3. METHODOLOGY

3.1 Research Approach

This research follows the **Design Science Research (DSR) methodology** (Hevner et al., 2004; Peffers et al., 2007), which is ideal for developing a practical AI artefact. The study will follow the six-step DSR process: problem identification, objective definition, design & development, demonstration, evaluation, and communication. A mixed approach combining **Design Science** (artefact creation) and **Experimental Research** (model testing and validation) will be used. The final deliverable will be a **cross-platform mobile application** built with React Native.

3.2 Technical Methods and Algorithms

- **Mentor–Student Matching**: TF-IDF vectorisation combined with cosine similarity on interests, degrees, and career goals.
- **Multi-Layered Verification**: Optical Character Recognition (OCR) for CV and certificate text extraction + rule-based keyword matching and confidence scoring.
- **Interview Scoring**: Structured questionnaire with keyword matching and basic sentiment analysis to determine verification strength.
- **Tiered Badge System**: Rule-based scoring logic to assign Bronze, Silver, or Gold badges based on combined OCR and interview results.
- **Real-Time Chat**: WebSocket-based bidirectional communication for secure messaging.
- **Privacy Mechanism**: Temporary encrypted storage with automatic file deletion after metadata extraction.

3.3 Tools and Platforms with Justification

Tool / Platform	Purpose	Justification
React Native + Expo	Frontend (Mobile App)	Chosen as the primary framework for developing a true cross-platform mobile app (Android + iOS) using a single codebase. Provides excellent support for camera/document upload, push notifications, real-time chat, and multilingual interface. Widely used in SLIIT final-year projects in 2025–2026 for fast development and native performance.
react-native-ocr / react-native-executorch	OCR Engine (on-device)	Lightweight, on-device OCR libraries that work directly on mobile phones without cloud dependency. Supports English and Sinhala text extraction from CVs/certificates, reduces privacy risks, and is completely free.
React Native Firebase	Real-time Chat & Storage	Official Firebase SDK for React Native. Enables fast real-time messaging, secure file upload, temporary encrypted storage, and automatic deletion rules. Free tier is sufficient and offers built-in security.
FastAPI (Python)	Backend APIs	High-performance RESTful APIs for matching logic, verification, and integration with the central AI platform. Easy integration with React Native and excellent documentation.
Supabase	Database & Authentication	Open-source PostgreSQL database with real-time capabilities and excellent React Native SDK. Free, secure, and fully privacy-compliant.
react-native-gifted-chat	Chat UI Component	Ready-to-use, highly customisable chat interface for React Native. Saves significant development time while providing a professional messaging experience with file sharing.

Table 1:Justification Table

React Native allows us to deliver a true mobile experience (push notifications for new messages, offline support, smooth document upload via camera/gallery) while keeping development time realistic for a final-year project. It also provides excellent multilingual support through react-i18next.

3.4 Data Collection and Ethical Considerations

- Synthetic datasets only (generated using Python Faker library – approximately 4,000 mentor and student profiles).
- Public Kaggle resume PDF datasets for OCR testing and HR Interview Questions dataset for interview scoring.
- No primary data will be collected from real students during the prototype phase.
- Full ethical approval will be obtained from the university ethics committee before any usability testing. Anonymity, informed consent, and strict data handling policies will be followed.

3.5 Validation Strategy and Performance Metrics

- **Verification Accuracy**: ≥ 85% (OCR extraction + interview scoring combined).
- **Matching Accuracy**: ≥ 85% (cosine similarity evaluation).
- **Usability & Satisfaction**: System Usability Scale (SUS) + Net Promoter Score (NPS) with target ≥ 85 from 30–50 test users.
- **Privacy Compliance**: Zero permanent storage of raw documents (verified through audit).
- **Real-Time Performance**: Message delivery reliability ≥ 95% and load time testing on real Android & iOS devices.
- Evaluation will be conducted with 30+ synthetic scenarios and a controlled user study (post-A/L students with consent).

4. HIGH-LEVEL SYSTEM ARCHITECTURE

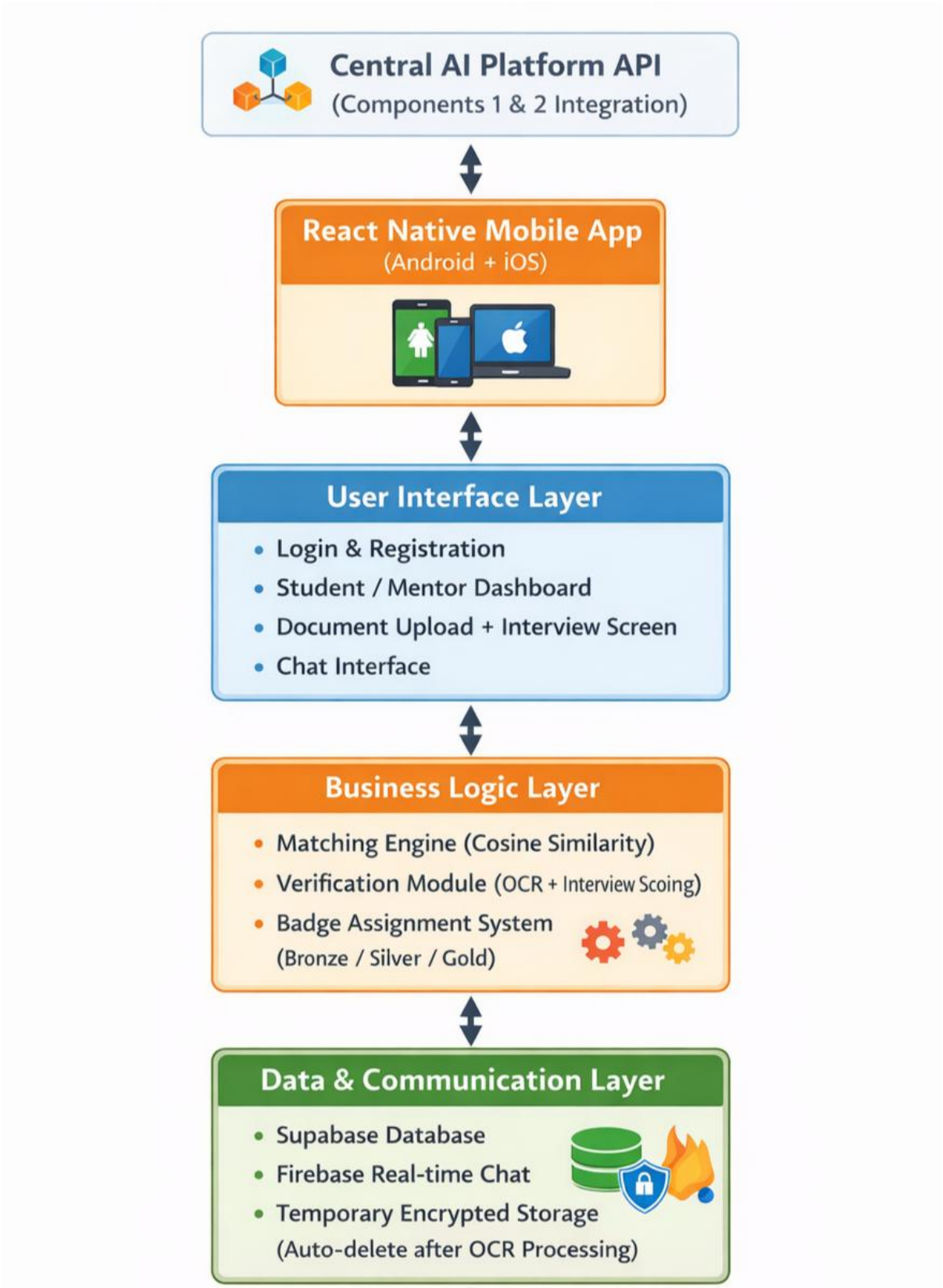


Figure 1:System Architecture Diagram

4.1 Individual Component and Their Parts

The Professional Mentorship and Alumni Network (Component 3) is the core focus of this proposal. It is clearly highlighted in the middle layers of the architecture diagram with the following key parts:

- **User Interface Layer** – Login & Registration, Student/Mentor Dashboard, Document Upload + Interview Screen, and Chat Interface.
- **Business Logic Layer** – Matching Engine (Cosine Similarity), Verification Module (OCR + Interview Scoring), and Badge Assignment System (Bronze/Silver/Gold).
- **Data & Communication Layer** – Supabase Database, Firebase Real-time Chat, and Temporary Encrypted Storage (with auto-deletion after OCR processing).

These parts work together to provide automated mentor verification, trustworthy matching, and secure communication.

4.2 Component Interactions

- The User Interface Layer sends user requests (document upload, interview answers, chat messages) to the Business Logic Layer.
- The Verification Module (OCR + Interview Scoring) interacts directly with the Badge Assignment System to assign Bronze, Silver, or Gold badges.
- Only mentors with verified badges are passed to the Matching Engine, which then recommends them to students.
- The Real-Time Chat module activates only after a successful match and badge verification.
- All layers communicate bidirectionally with the Central AI Platform API (Components 1 & 2) for seamless integration.

4.3 Data Flow Explanation

1. Student/Mentor opens the app and uploads CV/Certificates or completes the interview questionnaire.
2. On-device OCR processes the documents and extracts metadata; raw files are immediately deleted.
3. Interview scoring runs and the result is sent to the Badge Assignment System.
4. Verified profiles are stored in Supabase and fed into the Matching Engine.
5. When a match occurs, the Real-Time Chat module (Firebase) is activated for secure messaging.
6. All interactions are logged temporarily and deleted after processing to maintain privacy.

4.4 Scalability and Security Considerations

- **Scalability**: The architecture uses Firebase and Supabase, both of which support automatic real-time scaling for thousands of concurrent users. React Native ensures smooth performance even on mid-range mobile devices.

- **Security**: End-to-end encryption is applied to all chat messages. OCR processing happens on-device (no raw documents sent to the cloud). Temporary encrypted storage with automatic deletion policy is enforced. JWT authentication and full compliance with Sri Lanka's Personal Data Protection Act are implemented. The system is designed to be privacy-first while remaining highly scalable.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

- **Primary Stakeholders**
    - **Students** (undergraduates and post-A/L students) – who need reliable mentors for academic and career guidance.
    - **Mentors** (university alumni, faculty members, and industry professionals) – who provide guidance after successful verification.
- **Secondary Stakeholders**
    - **University Administration** (SLIIT and other institutions) – for oversight and integration with the central AI platform.
    - **System Administrators** – who monitor verification accuracy and overall system performance.
    - **End Users** (parents/guardians) – who may view limited progress reports with student consent.

5.2 Functional Requirements

1. The system shall allow students and mentors to register and create profiles with academic background, interests, and career goals.
2. The system shall enable users to upload CVs and certificates for mentor verification.
3. The system shall perform on-device Optical Character Recognition (OCR) on uploaded documents and extract relevant metadata.
4. The system shall present a structured interview questionnaire to mentors and automatically score responses.
5. The system shall assign Bronze, Silver, or Gold verified badges based on combined OCR and interview scores.
6. The system shall run a matching engine using cosine similarity and display recommended mentors/students.
7. The system shall provide a secure real-time chat interface between matched students and mentors.
8. The system shall allow users to view mentor profiles, badge levels, and availability.
9. The system shall integrate with the Central AI Platform (Components 1 & 2) to pull student data and share verified mentor information.
10. The system shall automatically delete raw uploaded documents after processing and metadata extraction.

5.3 Non-Functional Requirements

- **Security**: All chat messages shall use end-to-end encryption. Raw documents shall never be permanently stored (temporary encrypted storage with auto-deletion). The system shall comply with Sri Lanka's Personal Data Protection Act and implement JWT authentication.
- **Performance**: The mobile app shall load within 2 seconds on mid-range devices. Real-time chat messages shall be delivered with less than 1-second latency. OCR processing shall complete within 10 seconds on-device.
- **Usability**: The app shall support Sinhala, Tamil, and English languages with intuitive navigation. The System Usability Scale (SUS) score shall be ≥ 85.
- **Reliability & Availability**: The system shall achieve 99% uptime and support offline profile viewing.
- **Scalability**: The architecture shall support at least 5,000 concurrent users with automatic scaling via Firebase and Supabase.
- **Privacy**: No sensitive credentials or raw documents shall be stored permanently. Users shall have the right to request data deletion at any time.
- **Compatibility**: The mobile app shall run smoothly on Android 8.0+ and iOS 12+ devices.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market is Sri Lankan undergraduates and post-A/L students aged 18–25, focusing on those in IT, engineering, business, and related fields at institutions like SLIIT, University of Colombo, Moratuwa, and private campuses. Secondary markets include recent graduates and young professionals seeking mentorship, with potential expansion to school leavers and corporate employees.

Attribute	Description
Name	Tharindu Perera (or similar)
Age	21–24
Gender	Male / Female (balanced)
Education	Final-year undergrad or post-A/L (e.g., BSc IT, BEng)
Location	Urban/Semi-urban Sri Lanka (e.g., Negombo, Colombo, Gampaha)
Occupation	Student / Intern / Entry-level professional
Pain Points	Lack of trusted mentors, unverified advice, privacy concerns, mismatched career guidance
Goals	Find verified mentors, build networks, get career advice, improve employability
Tech Behavior	Mobile-first user (Android/iOS), active on WhatsApp/Instagram/LinkedIn, prefers simple apps with Sinhala/Tamil support
Willingness to Pay	Free basic access; LKR 500–1,500/month for premium (unlimited verified chats)

Table 2 :Sample Persona

6.2 Value Proposition

The platform offers **trusted, verified mentorship** in a secure mobile app, tailored for Sri Lankan students. It provides automated verification (OCR + interview + badges), privacy protection (auto-delete files), real-time chat, and AI integration — solving trust and accessibility issues that generic apps ignore.

6.3 Revenue Model

Freemium model:

- Free: Basic matching, limited chats.
- Subscription: LKR 750/month for unlimited verified mentors (Bronze+).
- Premium: LKR 1,500/month for priority Gold mentors + reports.
- Partnerships: University licensing (LKR 500,000/year).
- Future: Mentor commission on paid sessions (20%).

6.4 Cost Estimation

- Development (React Native tools): LKR 200,000.
- Cloud (Firebase/Supabase): LKR 100,000/year.
- Marketing: LKR 150,000/year.
- Legal/IP: LKR 75,000.
- Total First-Year: LKR 525,000.

6.5 Pricing Strategy

- Freemium entry to build user base.
- Monthly subscriptions with annual discounts (20%).
- Student discounts via university partnerships.

6.6 Competitive Advantage

Unique OCR + interview verification, privacy auto-delete, badge system, Sri Lankan customization, and mobile-first design — superior to LinkedIn or MentorcliQ.

6.7 Intellectual Property Considerations

Trademark the app name; copyright code; patent verification process; use open-source licenses for React Native/Firebase.

7. BUDGET AND JUSTIFICATION

Item	Estimated Cost (LKR)	Subtotal (LKR)
Hardware Costs	85,000 – 140,000	85,000 – 140,000
Software Licenses	0	0
Cloud Services	2,000 – 35,000	2,000 – 35,000
Human Effort Cost	0 – 180,000	0 – 180,000
Miscellaneous	15,000 – 37,000	15,000 – 37,000
Total Estimated Budget		**102,000 – 392,000**

Table 3:Budget and Justification Table

Hardware Costs (LKR 85,000 – 140,000)

- Mid-range Android test phone (LKR 45,000–65,000): Required to test the React Native app on real devices commonly used by Sri Lankan students.
- Necessary for camera-based document upload and on-device OCR accuracy, performance, battery usage, and heat during chat/verification sessions.
- Second-hand/refurbished iOS device (LKR 40,000–75,000): Essential for iOS compatibility testing.
- Expo/React Native requires a real iOS device for final builds, push notifications, and App Store preview.
- Real-device testing is critical — emulators cannot reliably simulate OCR, chat latency, or user experience on actual phones.

Software Licenses (LKR 0)

- React Native, Expo, FastAPI, EasyOCR, Faker, VS Code, Android Studio, and Xcode are all free and open-source.
- No commercial licenses or paid IDEs are required.
- This aligns with the open-source nature of the project and keeps costs at zero.

Cloud Services (LKR 2,000 – 35,000)

- Firebase (LKR 0–18,000/year): Free tier covers real-time chat, authentication, temporary file storage, and push notifications for the prototype phase.
- Supabase (LKR 0–12,000/year): Free tier sufficient for database (user profiles, matching data), authentication, and secure storage.
- Domain name (LKR 2,000–5,000/year): Optional low-cost .lk domain for sharing demo APK links or future deployment.
- These services are essential for real-time messaging and temporary encrypted storage — free tiers make costs minimal.

Human Effort Cost (LKR 0 – 180,000)

- Core development (React Native coding, OCR integration, backend APIs, chat UI, verification logic) is done by the student(s) — no external salary required (LKR 0).
- Optional freelancer for specific tasks (UI polish, chat customization, OCR tuning) — estimated LKR 80,000–180,000 if needed.
- Student-led effort is standard for final-year projects and keeps the cost low or zero.

Miscellaneous (LKR 15,000 – 37,000)

- Printing, binding, proposal documents, posters, and exhibition materials (LKR 5,000–12,000): Required for professional presentation and demo day.
- Testing incentives (snacks/vouchers for 30–50 beta testers) (LKR 10,000–25,000): Necessary for ethical usability testing (SUS surveys, feedback sessions) with fellow students.
- These ensure the project is presented professionally and evaluated ethically with real user input.

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Task Breakdown

Task ID	Task Description	Main Tasks	Responsible	Estimated Effort
1.0	Project Setup	Literature review, React Native + Expo setup, Firebase & Supabase setup	You	2 weeks
2.0	Backend Development	FastAPI APIs, Supabase database, matching logic	You	6–7 weeks
3.0	Verification Module	OCR document upload, interview questionnaire, scoring logic	You	5–6 weeks
4.0	Badge System	Bronze/Silver/Gold badge logic & display	You	3 weeks
5.0	Mobile App Frontend	Login, Dashboard, Document Upload, Interview Screen, Chat UI	You	7–8 weeks
6.0	Real-time Chat	Firebase chat implementation	You	4 weeks
7.0	Privacy & Integration	Auto-delete storage logic + Central AI Platform API integration	You	3 weeks
8.0	Testing & Validation	OCR accuracy, matching, chat testing, usability testing (SUS)	You	5 weeks
9.0	Documentation & Demo	Final report, poster, demo video, exhibition preparation	You	4 weeks

Figure 2:Work Breakdown Structure

8.2 Timeline Alignment

Month	Key Tasks	Milestone
Month 1	**Task 1.0**	**Environment ready**
Months 2–3	**Tasks 2.0 + 3.0**	**Backend + Verification ready**
Months 3–5	**Tasks 4.0 + 5.0 + 6.0**	**Full UI + Chat working**
Months 5–6	**Tasks 7.0 + 8.0 (first round)**	**Integrated prototype**
Months 6–7	**Task 8.0 (full testing)**	**Usability testing completed**
Month 8	**Task 9.0**	**Final report & exhibition ready**

Table 4:Timeline Alignment Table

8.3 Balanced Workload Distribution

Since this is an **individual project**, you will handle all tasks yourself.

- Weekly effort: **10–15 hours** (flexible around your job/studies).
- Focus strategy: First complete core features (Verification Module + Chat) → then UI polish → finally testing and documentation.
- Buffer time is included in each phase for revisions or unexpected delays.
- Workload is balanced by spreading heavy tasks (Frontend & Verification) across multiple months and keeping documentation for the last month.

9. GANTT CHART

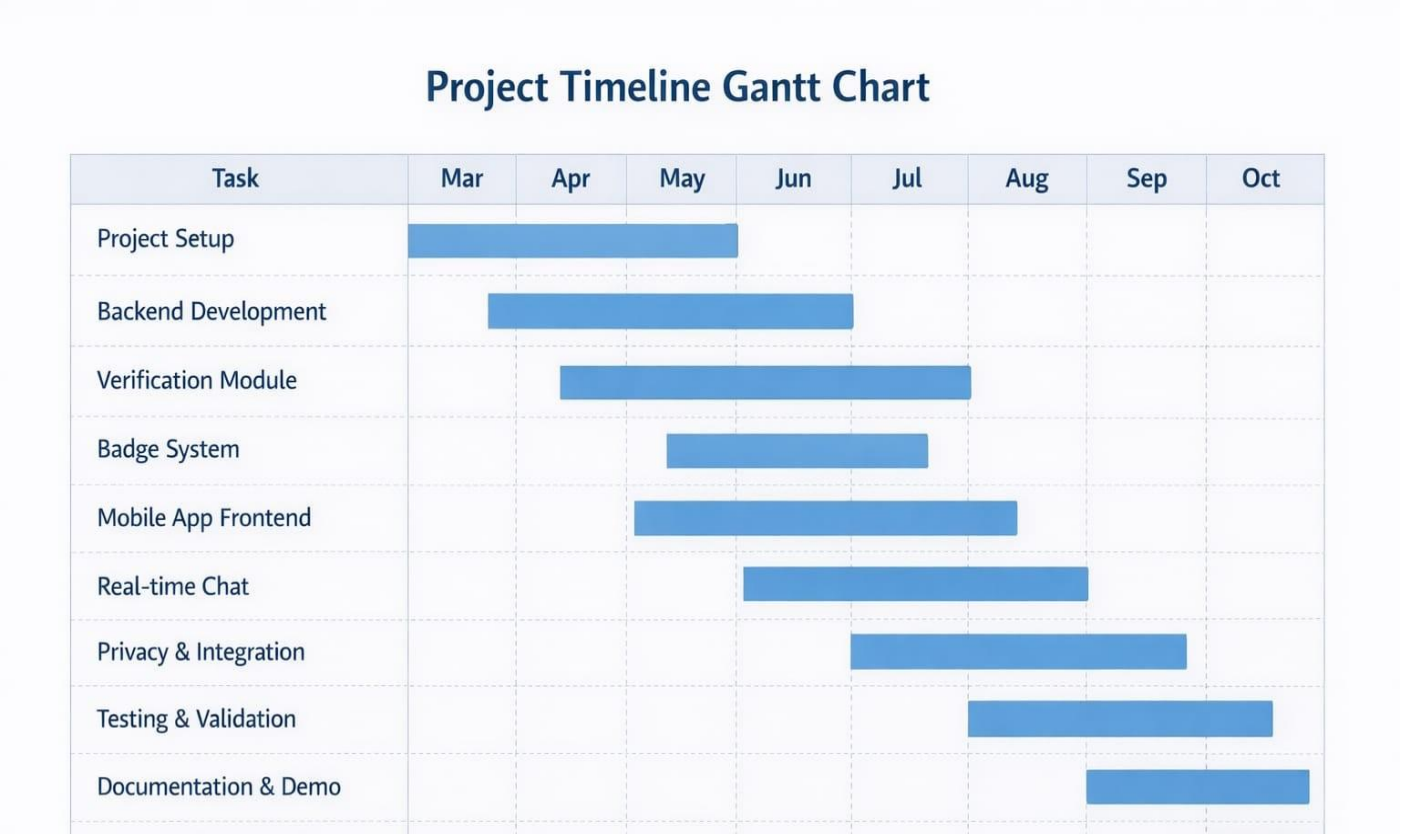


Figure 3:Gantt chart

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11.APPENDICES

Appendix A: Survey Instruments Usability Testing Questionnaire (System Usability Scale – SUS) (You will use this after demoing the app to 30–50 students/mentors)

1. I think that I would like to use this app frequently. (1 = Strongly Disagree, 5 = Strongly Agree)
2. I found the app unnecessarily complex. (1–5)
3. I found the app easy to use. (1–5)
4. I think I would need support from a technical person to use the app. (1–5)
5. I found the various functions in the app were well integrated. (1–5)
6. I thought there was too much inconsistency in the app. (1–5)
7. I would imagine that most people would learn to use the app very quickly. (1–5)
8. I found the app very cumbersome to use. (1–5)
9. I felt very confident using the app. (1–5)
10. I needed to learn a lot of things before I could get going with the app. (1–5)

Additional Questions

- How trustworthy did the mentor badges (Bronze/Silver/Gold) look? (1–5)
- Was the document upload and verification process easy to understand? (1–5 + comment)
- Did you feel your privacy was protected when uploading documents? (1–5 + comment)
- Any suggestions for improving the chat or interview part?

Appendix B: Detailed Diagrams

1. **Verification Data Flow Diagram**

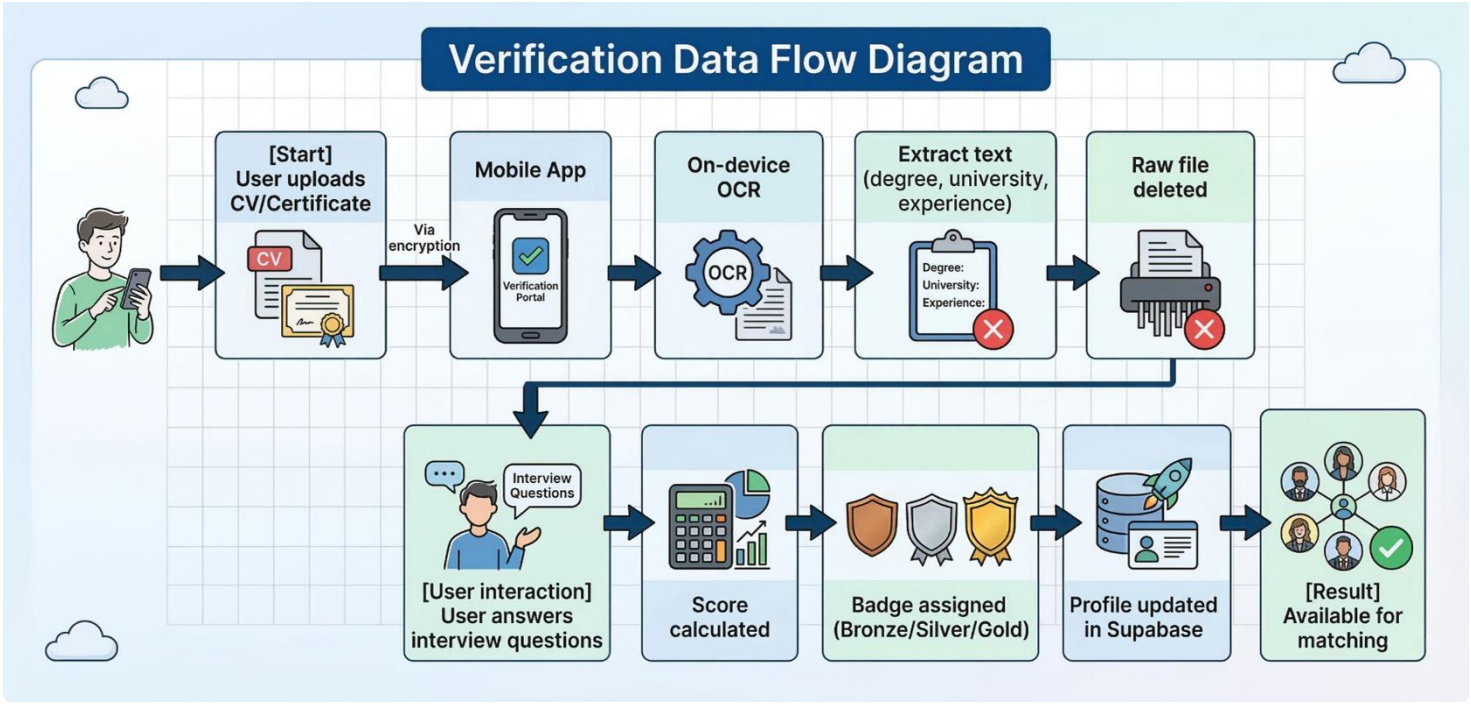


Figure 4.1: Verification Data Flow Diagram

2. Mentorship Platform Relationship Diagram

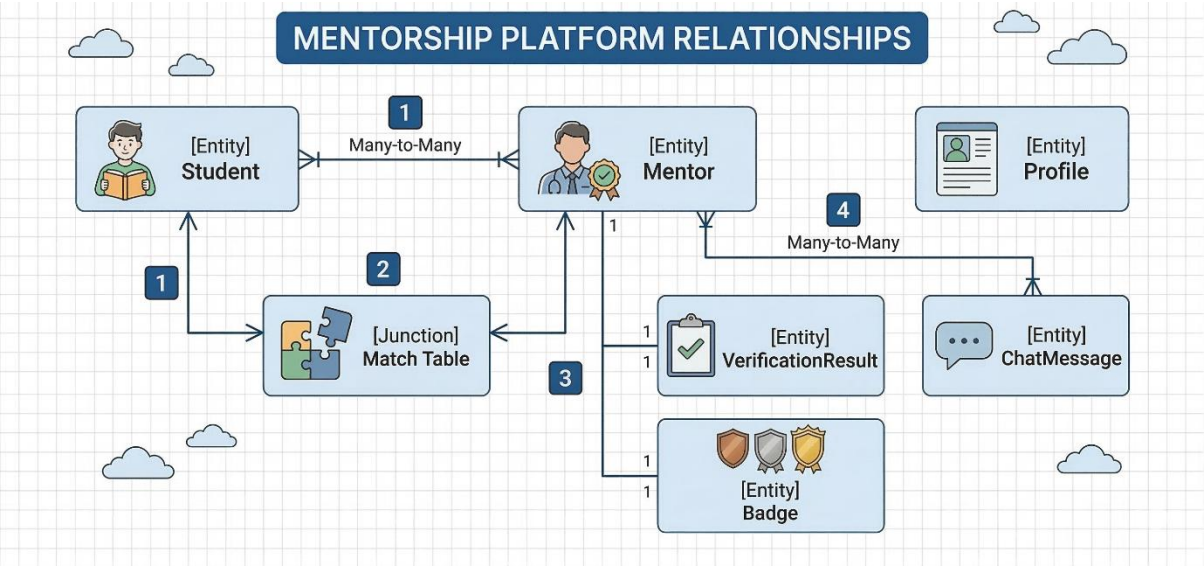


Figure 5.2: Mentorship Platform Relationship Diagram

Appendix C: Preliminary UI Mockups

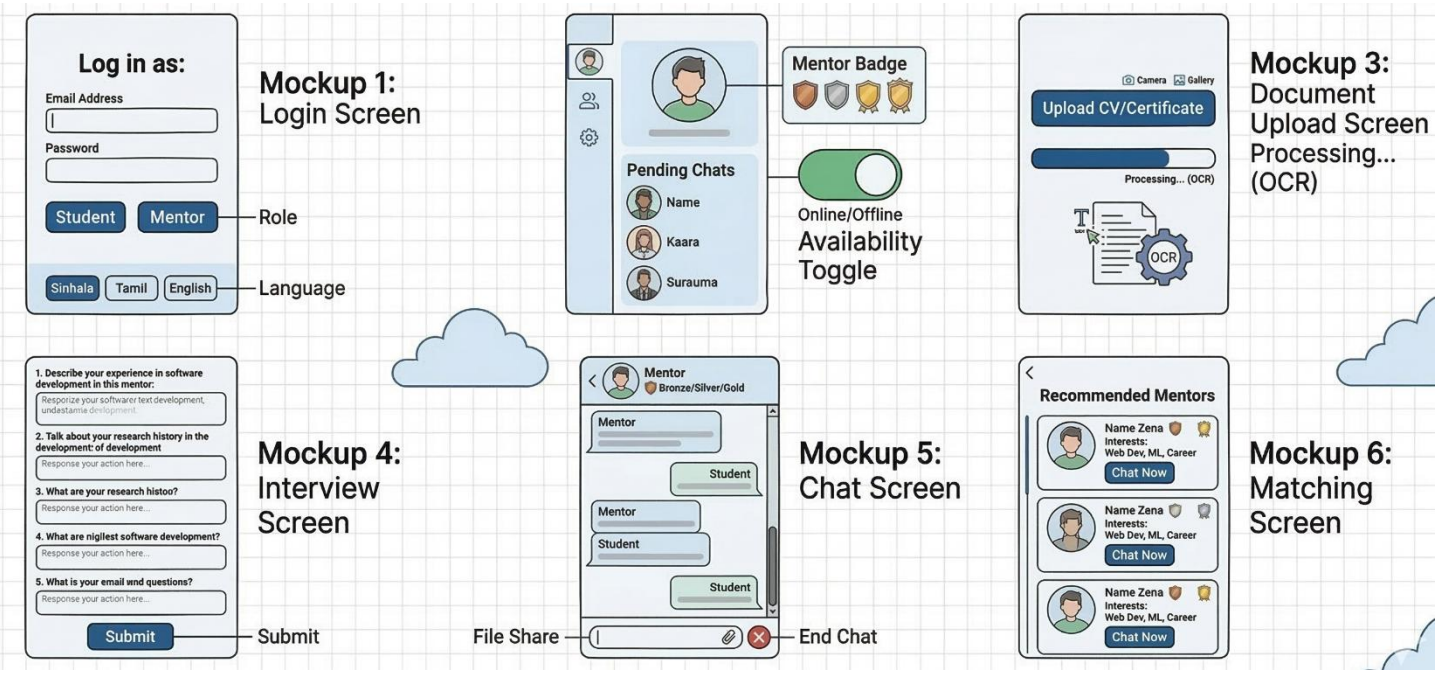


Figure 6.3: Preliminary UI Mockups

Appendix D: Risk Analysis

Risk ID	Risk Description	Probability	Impact	Mitigation Plan
R1	OCR fails on poor-quality or handwritten documents	Medium	High	Use high-quality sample images for testing; add fallback manual upload option; display clear instructions. .
R2	Low number of beta testers for usability study	Medium	High	Recruit early from university WhatsApp groups; offer small incentives (snacks/vouchers).
R3	Firebase/Supabase free tier limits exceeded	Low	Medium	Monitor usage weekly; switch to paid tier if needed (budget has buffer).
R4	Privacy concern from users about document upload	Medium	Very High	Show clear "Files deleted after processing" message; get ethics approval; use on-device OCR.
R5	Delay due to part-time job or illness	Medium	Medium	Build 2-week buffer in timeline; prioritize core features (verification + chat) first.

Figure 7.4:Risk Analysis Table

Appendix E: Ethical Documents

Ethics Clearance Plan

- Submit ethics application to SLIIT Research Ethics Committee before any user testing.
- All testing will use informed consent (see sample below).
- No real personal data collected during prototype — only synthetic profiles used.
- Raw documents deleted immediately after OCR/metadata extraction.
- Participants can withdraw at any time; all feedback anonymized.